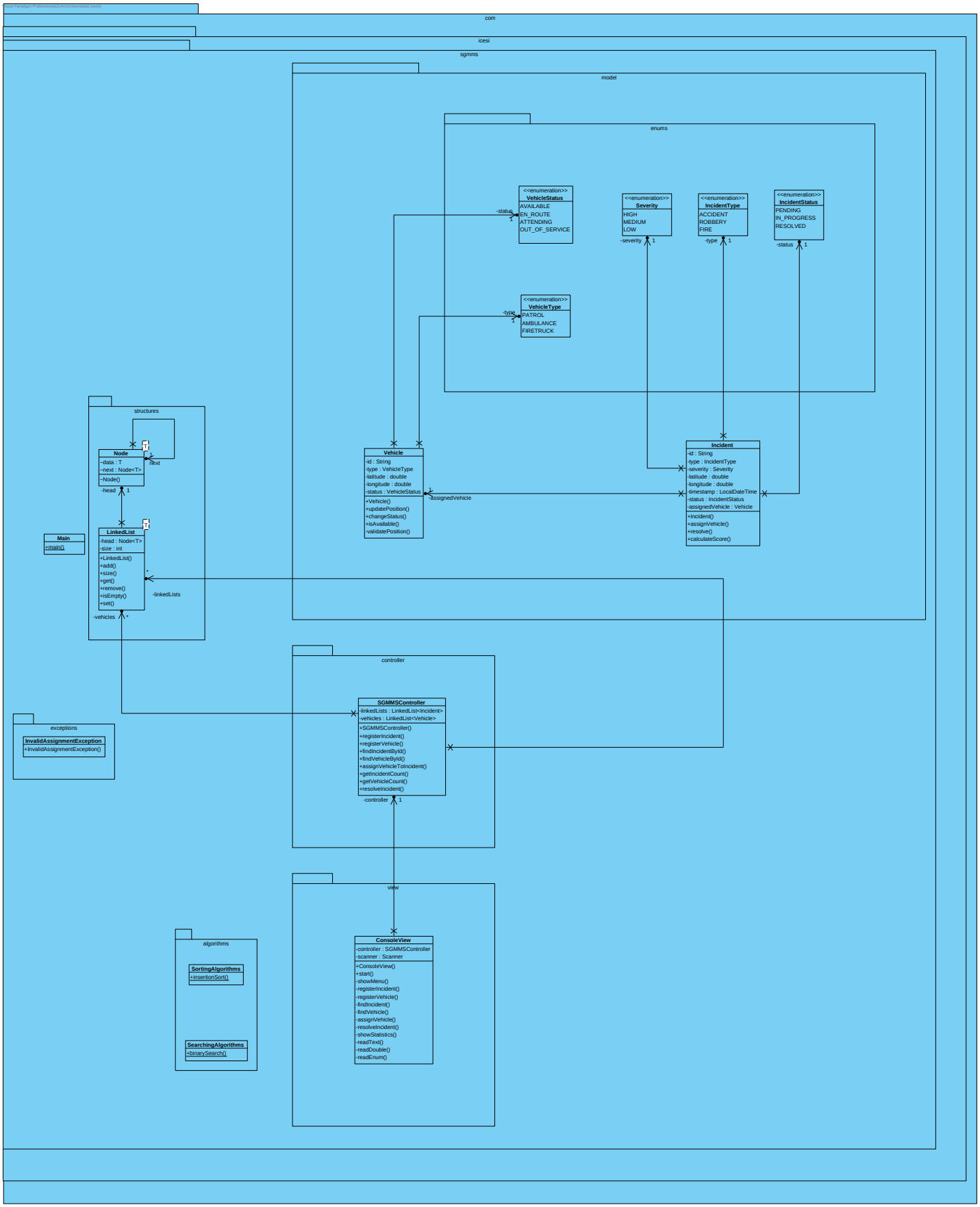
com
icesi
sgmms
model
enums
<<enumeration>>
VehicleStatus
AVAILABLE
EN_ROUTE
ATTENDING
OUT_OF_SERVICE
<<enumeration>>
Severity
HIGH
MEDIUM
LOW
<<enumeration>>
IncidentType
ACCIDENT
ROBBERY
FIRE
<<enumeration>>
IncidentStatus
PENDING
IN_PROGRESS
RESOLVED
<<enumeration>>
VehicleType
PATROL
AMBULANCE
FIRETRUCK
structures
Node
-data : T
-next : Node<T>
-Node()
LinkedList
-head : Node<T>
-size : int
+LinkedList()
+add()
+size()
+get()
+remove()
+isEmpty()
+set()
Main
+main()
Vehicle
-id : String
-type : VehicleType
-latitude : double
-longitude : double
-status : VehicleStatus
+Vehicle()
+updatePosition()
+changeStatus()
+isAvailable()
-validatePosition()
Incident
-id : String
-type : IncidentType
-severity : Severity
-latitude : double
-longitude : double
-timestamp : LocalDateTime
-status : IncidentStatus
-assignedVehicle : Vehicle
+Incident()
+assignVehicle()
+resolve()
+calculateScore()
controller
SGMMSController
-linkedLists : LinkedList<Incident>
-vehicles : LinkedList<Vehicle>
+SGMMSController()
+registerIncident()
+registerVehicle()
+findIncidentById()
+findVehicleById()
+assignVehicleToIncident()
+getIncidentCount()
+getVehicleCount()
+resolveIncident()
exceptions
InvalidAssignmentException
+InvalidAssignmentException()
view
ConsoleView
-controller : SGMMSController
-scanner : Scanner
+ConsoleView()
+start()
-showMenu()
-registerIncident()
-registerVehicle()
-findIncident()
-findVehicle()
-assignVehicle()
-resolveIncident()
-showStatistics()
-readText()
-readDouble()
-readEnum()
algorithms
SortingAlgorithms
+insertionSort()
SearchingAlgorithms
+binarySearch()